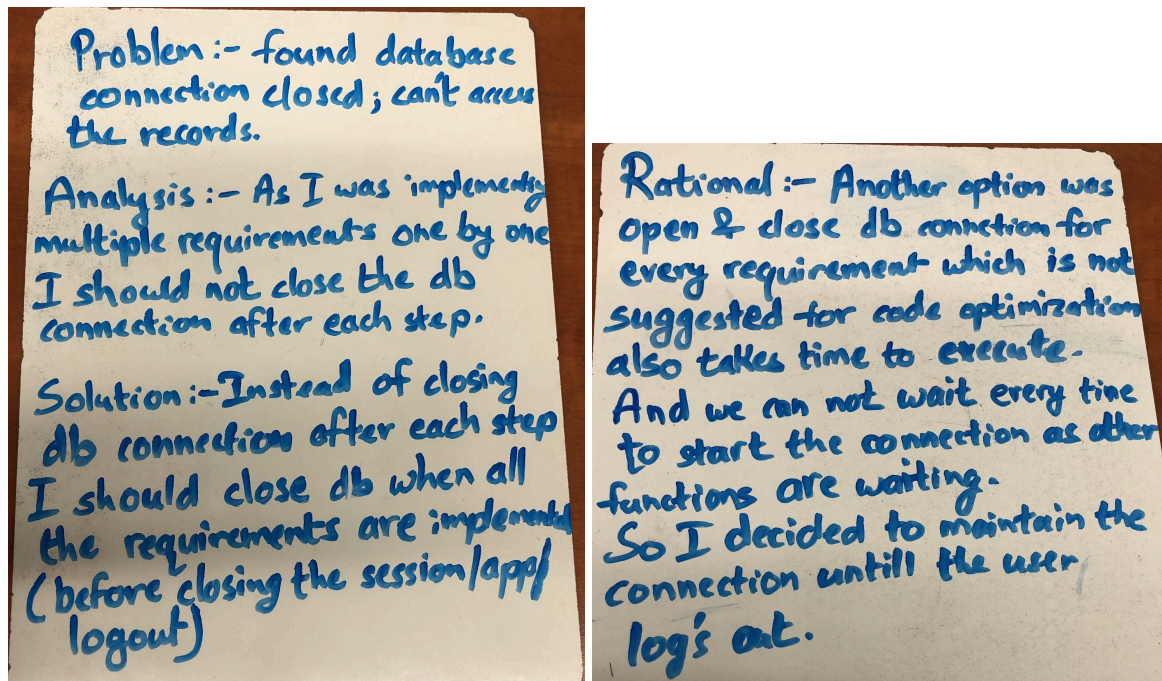


Faced a problem and how to solve it. (white board activity)



## Database Connection Handling Issue

### Problem Statement:

While executing multiple functions sequentially, each function independently opened a database session and closed it upon completion. As a result, once the first function closed the connection, subsequent functions attempted to access the database but failed because the connection was no longer active. This led to recurring errors such as "Can't access the database as the connection is closed."

### Root Cause:

- Each function was managing its own session lifecycle.
- Closing the session at the end of the first function caused subsequent functions to operate on a closed session.
- There was no shared or persistent session maintained across the series of function calls.

### Solution Implemented:

To resolve the issue, the session lifecycle was adjusted:

- Maintain a single database session for the duration of the workflow (i.e., across multiple function calls).
- Only close the session once all dependent functions have executed successfully.
- This ensures all functions share a valid, active session without encountering premature disconnection errors.

**Best Practice Recommendation:**

- Use a context manager or dependency injection pattern to manage the session lifecycle more predictably.
- Clearly define session boundaries at the workflow level (not the function level) when multiple functions must share the same transaction or connection.
- Ensure proper cleanup by closing the session at the end of the workflow to avoid resource leaks.